

analysis_REM_phasic_tonic(1)

December 15, 2025

```
[1]: import mne
import numpy as np
import matplotlib.pyplot as plt
from glob import glob
from pathlib import Path
import os
from scipy.signal import find_peaks
from scipy.signal import welch
```

```
[2]: import warnings
warnings.filterwarnings("ignore", category=RuntimeWarning)

import mne
mne.set_log_level("ERROR")
```

```
[3]: file_path= Path("/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/
↳sleep-cassette")
```

0.0.1 Data loading

```
[4]: psg_file=sorted(glob(os.path.join(file_path, "*PSG.edf")))
hyp_file=sorted(glob(os.path.join(file_path, "*Hypnogram.edf")))
psg_file
```

```
[4]: ['/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
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[illegible]

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[5]: hyp_file

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'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4591GY-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4592GY-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4601EC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4602EJ-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4611EG-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4612EA-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4621EV-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4622EJ-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4631EM-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4632EA-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4641EP-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4642EP-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4651EP-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4652EG-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4661EJ-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4662EJ-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4671GJ-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4672GV-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4701EC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4702EA-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4711EC-Hypnogram.edf',

```

'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4712EA-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4721EC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4722EM-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4731EM-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4732EJ-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4741EA-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4742EC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4751EC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4752EM-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4761EP-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4762EG-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4771GC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4772GC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4801GC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4802GV-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4811GG-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4812GV-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4821GC-Hypnogram.edf',
'/home/fateme/Downloads/sleep-edf-database-expanded-1.0.0/sleep-
cassette/SC4822GC-Hypnogram.edf']

```

0.0.2 data exploration

```

[5]: raw=mne.io.read_raw_edf(psg_file[0])
raw.info

```

```

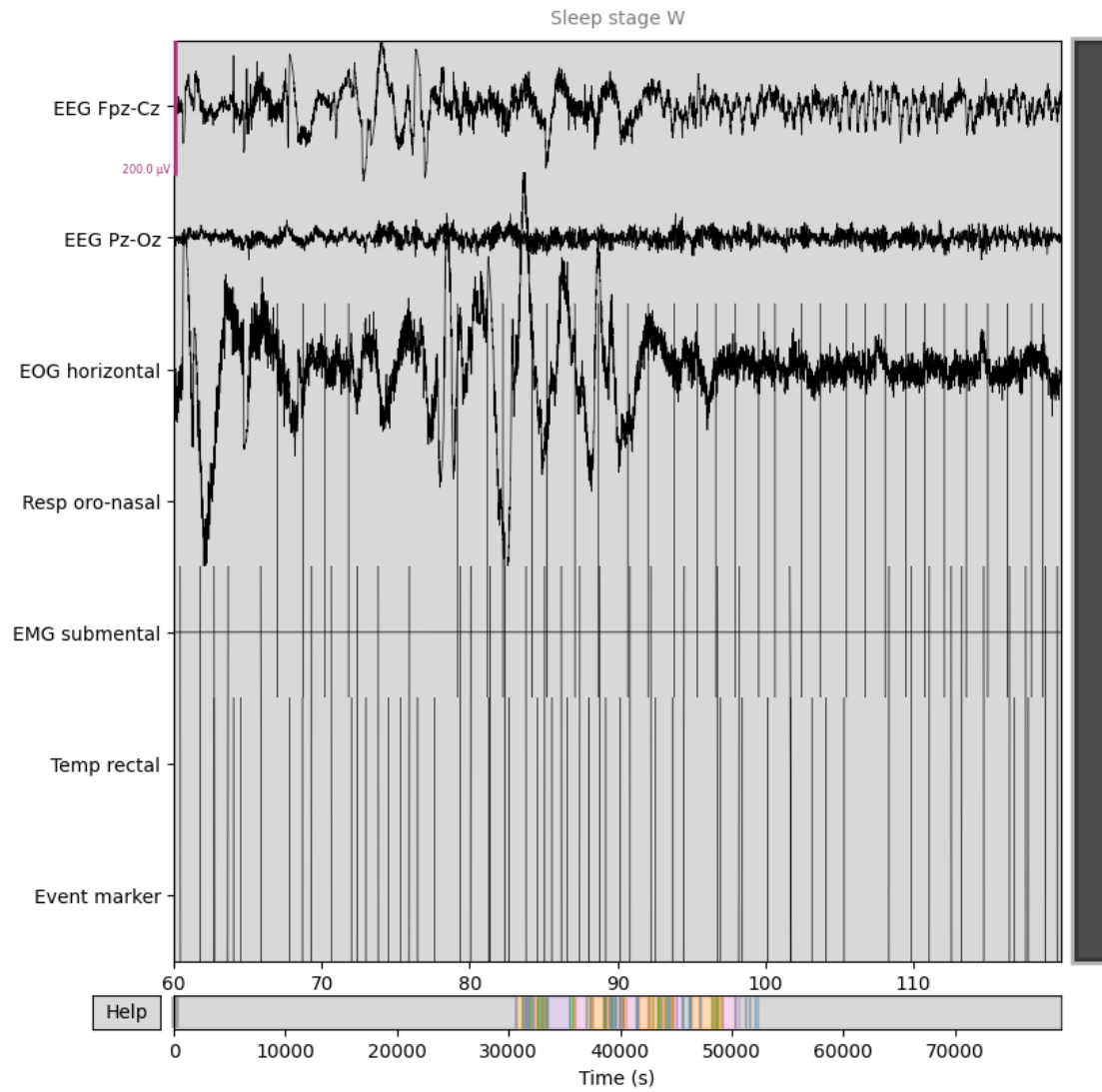
[5]: <Info | 8 non-empty values
      bads: []
      ch_names: EEG Fpz-Cz, EEG Pz-Oz, EOG horizontal, Resp oro-nasal, EMG ...

```

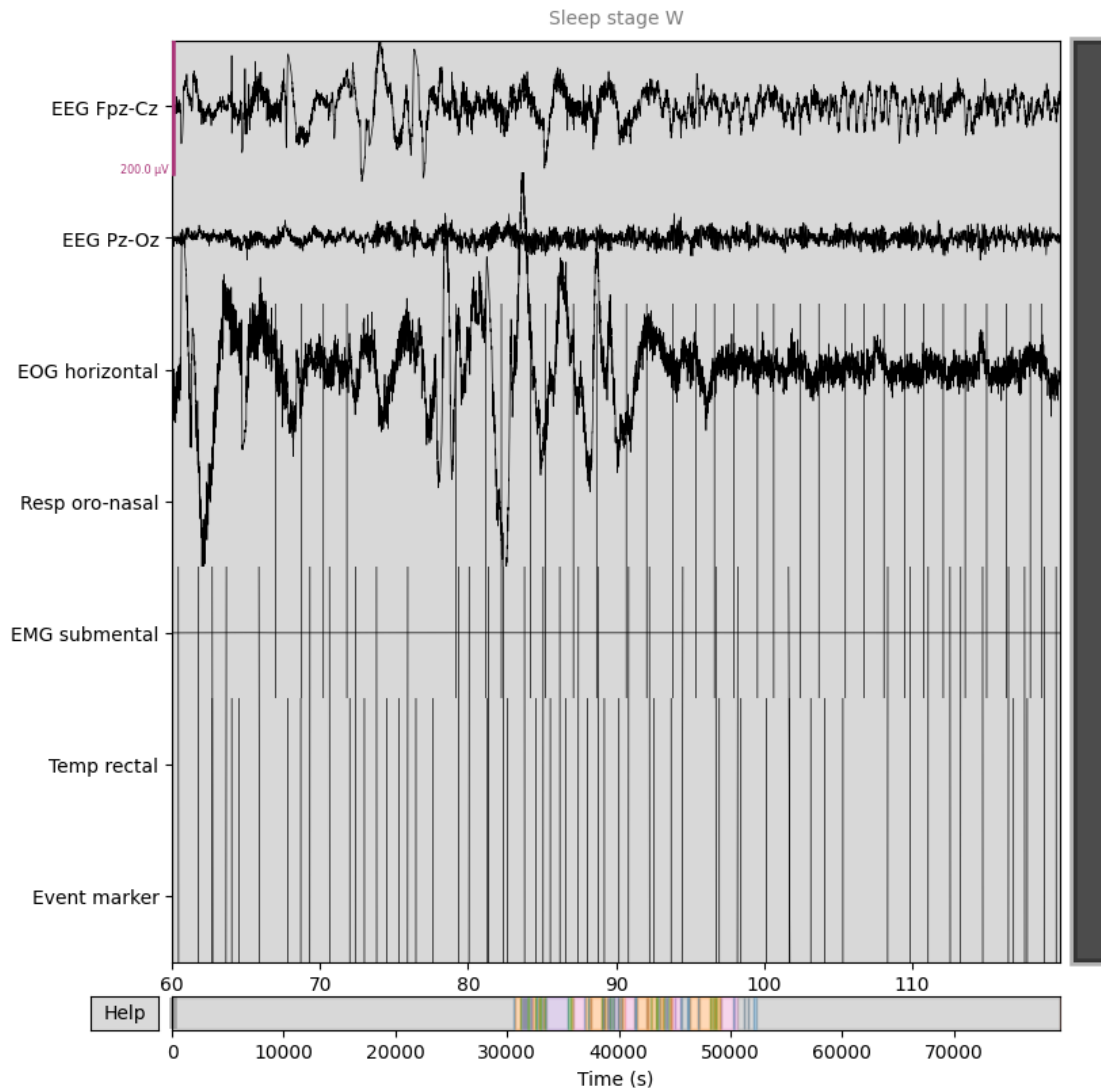
```
chs: 7 EEG
custom_ref_applied: False
highpass: 0.0 Hz
lowpass: 50.0 Hz
meas_date: 1989-04-24 16:13:00 UTC
nchan: 7
projs: []
sfreq: 100.0 Hz
subject_info: <subject_info | his_id: X, sex: 2, first_name: Female, last_name:
33yr>
>
```

```
[6]: raw = mne.io.read_raw_edf(psg_file[0], preload=True)
annot = mne.read_annotations(hyp_file[0])
raw.set_annotations(annot, emit_warning=False)

raw.plot(
    start=60,
    duration=60,
    scalings=dict(
        eeg=1e-4,
        resp=1e3,
        eog=1e-4,
        emg=1e-7,
        misc=1e-1
    )
)
```



[6] :



0.0.3 pre processing

```
[7]: def preprocess(psg_path, hyp_path, EEG_CH, EOG_CH):
    raw = mne.io.read_raw_edf(psg_path, preload=True, verbose=False)
    raw.set_annotations(mne.read_annotations(hyp_path))

    sf = raw.info['sfreq']

    eeg_raw = raw.copy().pick([EEG_CH])
    eeg_raw.filter(0.5, 45, verbose=False)
    eeg = eeg_raw.get_data()[0] * 1e6
```

```
eog_raw = raw.copy().pick([EOG_CH])
eog = eog_raw.get_data()[0] * 1e6

return raw, eeg, eog, sf
```

```
[7]: annot.description
```

```
[7]: array(['Sleep stage W', 'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 3',
'Sleep stage 4', 'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 3',
'Sleep stage 4', 'Sleep stage W', 'Sleep stage 3', 'Sleep stage 2',
'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2',
'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 1', 'Sleep stage 3',
'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage R', 'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 1', 'Sleep stage 2',
'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 1', 'Sleep stage 2',
'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 1', 'Sleep stage 2',
'Sleep stage 1', 'Sleep stage 2', 'Sleep stage R', 'Sleep stage 2',
'Sleep stage 1', 'Sleep stage W', 'Sleep stage 1', 'Sleep stage 2',
'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 1', 'Sleep stage 2',
'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2',
'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 4',
'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2',
'Sleep stage 4', 'Sleep stage 3', 'Sleep stage 4', 'Sleep stage 3',
'Sleep stage 4', 'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 4', 'Sleep stage 2', 'Sleep stage R', 'Sleep stage 1',
'Sleep stage R', 'Sleep stage W', 'Sleep stage 1', 'Sleep stage W',
'Sleep stage 1', 'Sleep stage W', 'Sleep stage 1', 'Sleep stage 2',
'Sleep stage W', 'Sleep stage 1', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 2', 'Sleep stage 3', 'Sleep stage 2', 'Sleep stage 3',
'Sleep stage 2', 'Sleep stage 1', 'Sleep stage R', 'Sleep stage W',
'Sleep stage 1', 'Sleep stage R', 'Sleep stage W', 'Sleep stage 1',
'Sleep stage W', 'Sleep stage 1', 'Sleep stage W', 'Sleep stage 1',
'Sleep stage W', 'Sleep stage ?'], dtype='<U13')
```

0.0.4 plot hypnogram

```
[8]: def plot_hypnogram_with_rem(raw, annotations, filename, subject_id):

    fs = raw.info["sfreq"]
    t_end = raw.times[-1]

    stage_map = {
        "Sleep stage W": 0,
        "Sleep stage 1": 1,
        "Sleep stage 2": 2,
        "Sleep stage 3": 3,
        "Sleep stage 4": 4,
        "Sleep stage R": 5,
    }

    x_all, y_all = [], []
    rem_segments = []

    for o, d, s in zip(annotations.onset, annotations.duration, annotations.
↳description):

        label = s.strip()
        start = max(o, 0)
        end = min(o + d, t_end)

        stage = stage_map.get(label, np.nan)
        if np.isnan(stage):
            continue

        x_all += [start, end]
        y_all += [stage, stage]

        if label == "Sleep stage R":
            rem_segments.append((start, end, stage))

    plt.figure(figsize=(12, 4))

    plt.step(x_all, y_all, where="post", color="blue", linewidth=1)

    for start, end, stage in rem_segments:
        plt.hlines(
            y=stage,
            xmin=start,
            xmax=end,
            color="black",
            linewidth=4
```

```

    )

    plt.xlim(0, t_end)
    plt.ylim(-0.5, 5.5)
    plt.yticks([0, 1, 2, 3, 4, 5], ["W", "N1", "N2", "N3", "N4", "R"])
    plt.xlabel("Time (s)")
    plt.ylabel("Sleep stage")
    plt.title(f"Hypnogram (REM highlighted) - {filename}")
    plt.tight_layout()
    plt.savefig(f"hypnogram_subject_{subject_id}.png", dpi=300,
    ↪ bbox_inches='tight')
    plt.show()

N_subjects = 5

for i, (psg_path, hyp_path) in enumerate(zip(psg_file[:N_subjects], hyp_file[:
    ↪ N_subjects]), start=1):

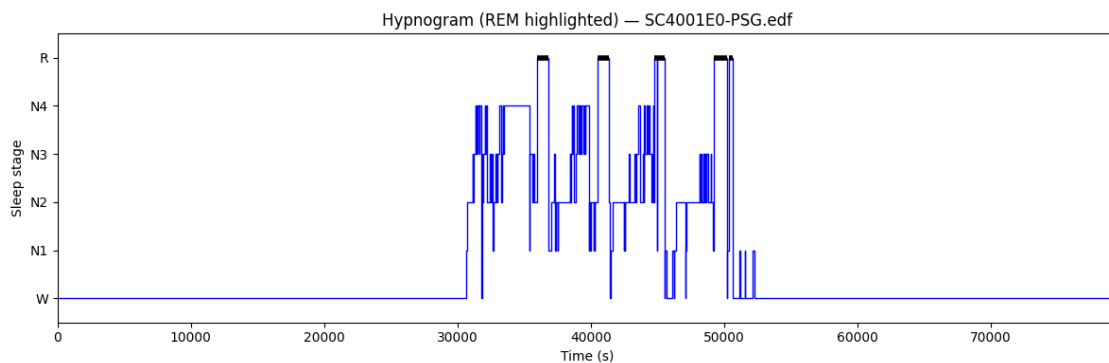
    print(f"Processing {os.path.basename(psg_path)} ...")

    raw = mne.io.read_raw_edf(psg_path, preload=True, verbose=False)
    annotations = mne.read_annotations(hyp_path)
    raw.set_annotations(annotations)

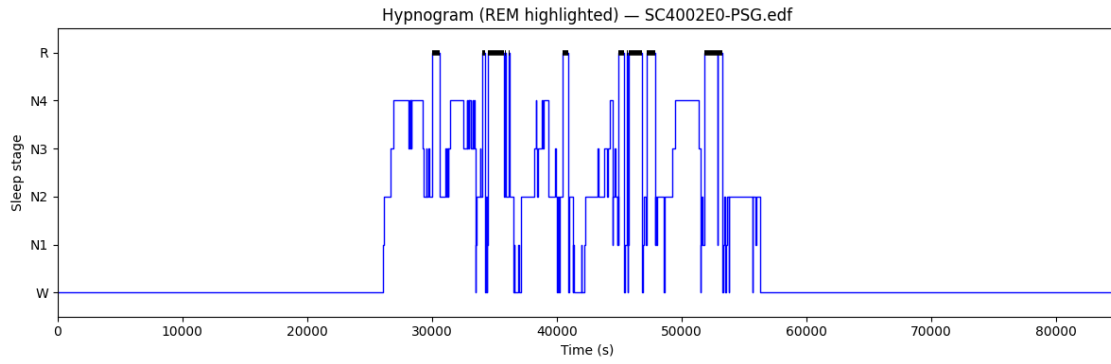
    plot_hypnogram_with_rem(raw, annotations, os.path.basename(psg_path), i)

```

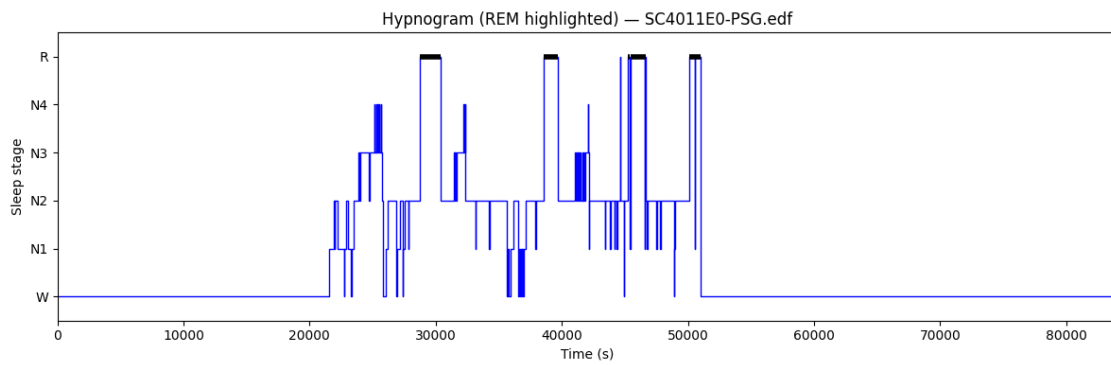
Processing SC4001E0-PSG.edf ...



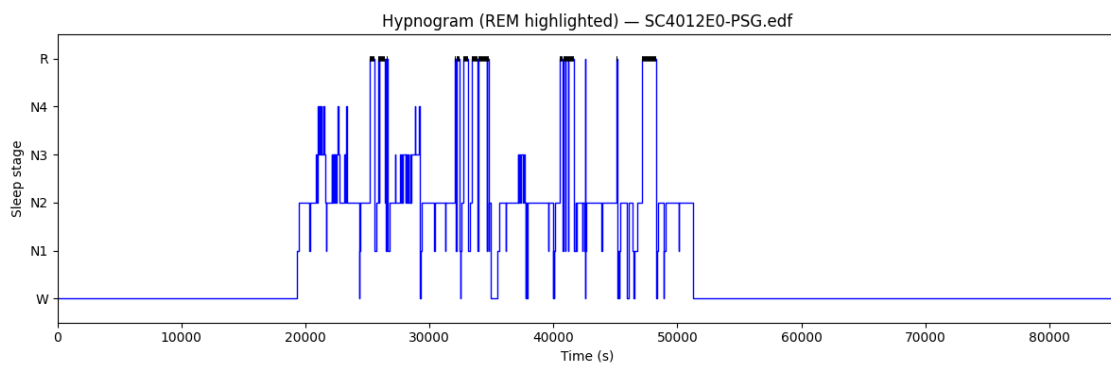
Processing SC4002E0-PSG.edf ...



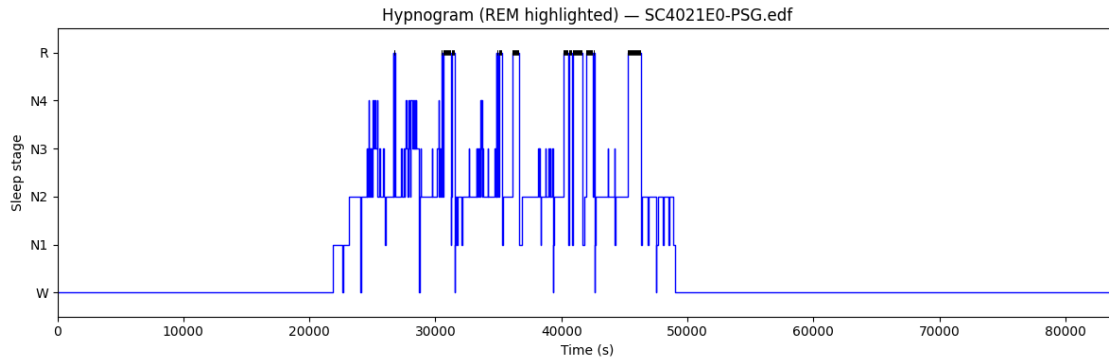
Processing SC4011E0-PSG.edf ...



Processing SC4012E0-PSG.edf ...



Processing SC4021E0-PSG.edf ...



0.0.5 detect phasic and tonic on EOG

0.0.6 set variable from paper

```
[9]: window_sec = 4
min_gap_sec = 8
phasic_uv = 100 # threshold for phasic
tonic_uv = 25 # for tonic
EOG_CH = 'EOG horizontal'
EEG_CH = 'EEG Pz-Oz'
N_subjects= 20
```

```
[10]: # get a boolean mask of the REMs in data
def rem_mask(raw, rem_label="Sleep stage R"):
    n_samples = raw.n_times
    mask = np.zeros(n_samples, dtype=bool)
    for o, d, s in zip(raw.annotations.onset, raw.annotations.duration, raw.
↳ annotations.description):
        if s == rem_label:
            start, end = raw.time_as_index([o, o + d])

            mask[start:end] = True
    return mask
```

```
[11]: def detect_phasic_tonic(eog_data, sf,
↳ rem_mask, window_sec=4, min_gap_sec=8, phasic_uv=100, tonic_uv=25, min_peak_distance_sec=0.
↳ 2):

    win = int(window_sec * sf)
    gap = int(min_gap_sec * sf)
    min_dist = int(min_peak_distance_sec * sf)

    phasic = []
    tonic = []
```

```

last_end = -gap
n = len(eog_data)

for start in range(0, n - win, win):
    end = start + win

    # only inside REM
    if not rem_mask[start:end].all():
        continue

    # 8 seconds gap
    if start - last_end < gap:
        continue

    seg = eog_data[start:end]
    abs_seg = np.abs(seg)

    # tonic rule
    if abs_seg.max() < tonic_uv:
        tonic.append((start, end))
        last_end = end
        continue

    # detect real peaks >= 100  $\mu$ V
    """
    peaks, info = find_peaks(abs_seg,
                             height=phasic_uv,
                             distance=min_dist)
    """

    peaks, info = find_peaks(abs_seg, height=phasic_uv, distance=min_dist,
    ↪prominence=50)

    #at least two consecutive eye movement
    if len(peaks) >= 2:
        phasic.append((start, end))
        last_end = end

    return phasic, tonic

```

```

[12]: N_subjects=20
for i, (psg_path, hyp_path) in enumerate(zip(psg_file[:N_subjects],hyp_file[:
    ↪N_subjects]), start=1):

    print(f"[{i}]  {os.path.basename(psg_path)} ...")

```

```

raw, eeg, eog, sf = preprocess(psg_path, hyp_path, EEG_CH, EOG_CH)

rem_mask_bool = rem_mask(raw, rem_label="Sleep stage R")

phasic, tonic = detect_phasic_tonic(eog, sf, rem_mask_bool)

# detect phasic and tonic windows
phasic, tonic = detect_phasic_tonic(eog, sf, rem_mask_bool)
print(f"  phasic windows: {len(phasic)}, tonic windows: {len(tonic)}")

if not phasic and not tonic:
    print("  no phasic or tonic windows to plot")
    continue

# choose first phasic window, otherwise first tonic
if phasic:
    first_start, first_end = phasic[0]
else:
    first_start, first_end = tonic[0]

padding    = int(30 * sf)
start_samp = max(first_start - padding, 0)
end_samp   = min(first_end + padding, len(eog))

times = np.arange(start_samp, end_samp) / sf
seg    = eog[start_samp:end_samp]

plt.figure(figsize=(10, 4))
plt.plot(times, seg, linewidth=0.8, label="EOG")

# highlight phasic windows inside this segment (blue)
added_phasic = False
for s, e in phasic:
    if e <= start_samp or s >= end_samp:
        continue
    plt.axvspan(
        s / sf, e / sf,
        alpha=0.3,
        color="blue",
        label="phasic" if not added_phasic else None,
    )
    added_phasic = True

added_tonic = False

```



```

for s, e in tonic:
    if e <= start_samp or s >= end_samp:
        continue
    plt.axvspan(
        s / sf, e / sf,
        alpha=0.2,
        color="green",
        label="tonic" if not added_tonic else None,
    )
    added_tonic = True

#seg_uv = seg * 1e6
big_idx = np.where(np.abs(seg) >= phasic_uv)[0]
#big_idx = np.where(np.abs(seg_uv) >= phasic_uv)[0]

"""
if len(big_idx) > 0:
    plt.scatter(
        times[big_idx],
        seg[big_idx],
        s=10,
        marker="o",
        label=f"|EOG| {phasic_uv}  $\mu$ V",
    )
"""

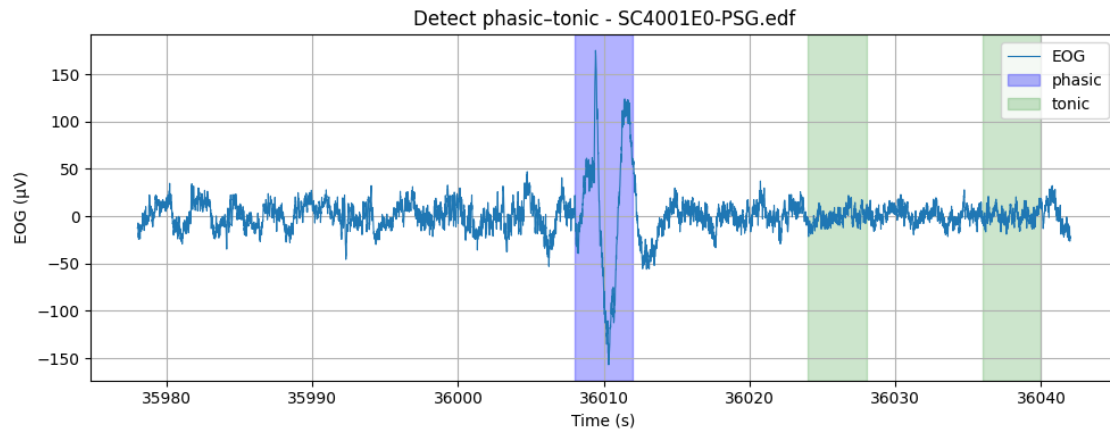
plt.title("Detect phasic-tonic - " + os.path.basename(psg_path))
plt.xlabel("Time (s)")
plt.ylabel("EOG ( $\mu$ V)")
plt.legend(loc="upper right")
plt.tight_layout()
plt.grid()
plt.savefig(f"EOG_phasic_tonic_subject_{i}.png", dpi=300,
            bbox_inches='tight')
plt.show()

```

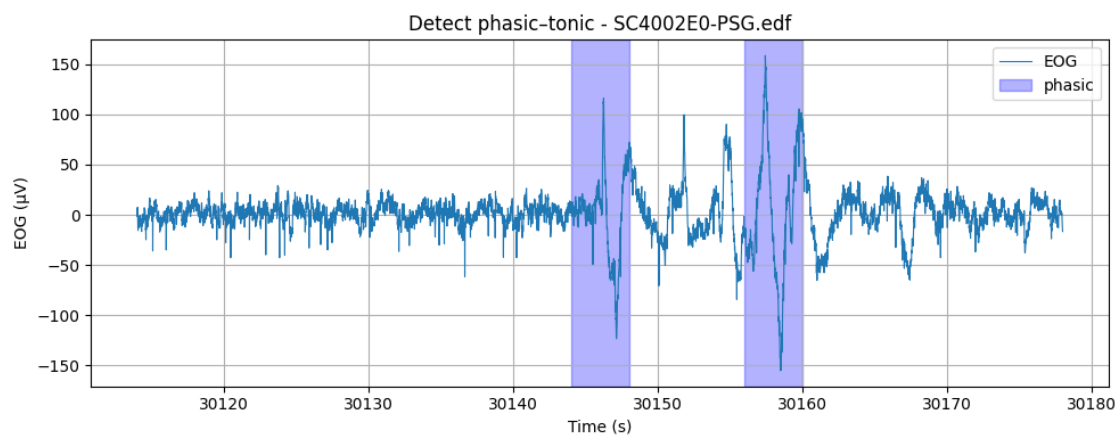
```

[1] SC4001E0-PSG.edf ...
    phasic windows: 60, tonic windows: 71

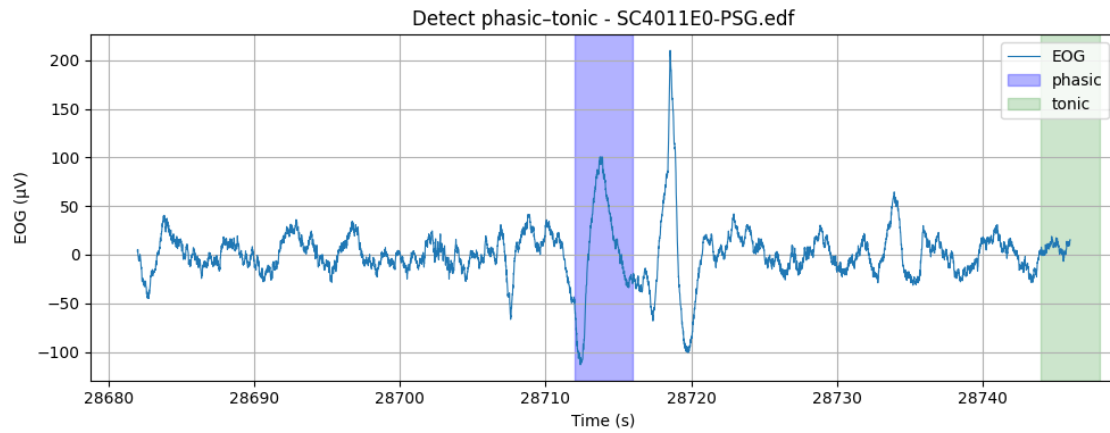
```



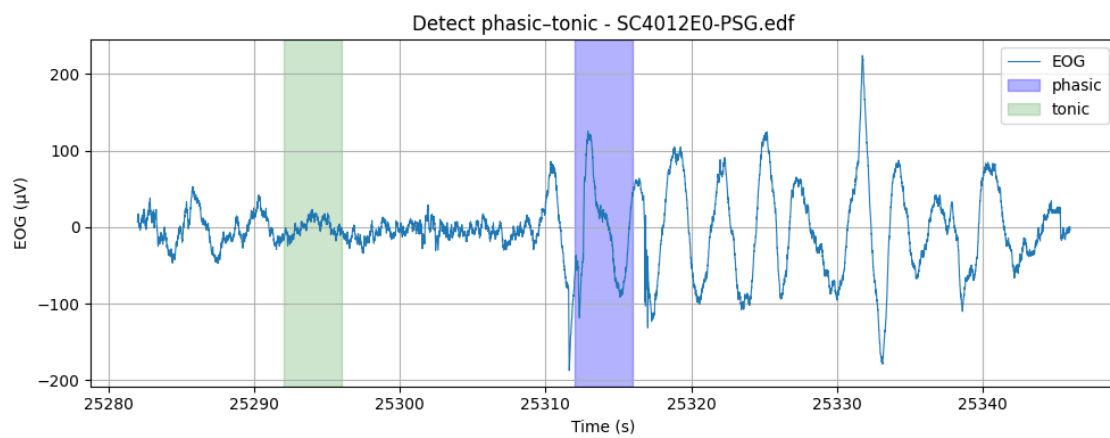
[2] SC4002E0-PSG.edf ...
 phasic windows: 89, tonic windows: 12



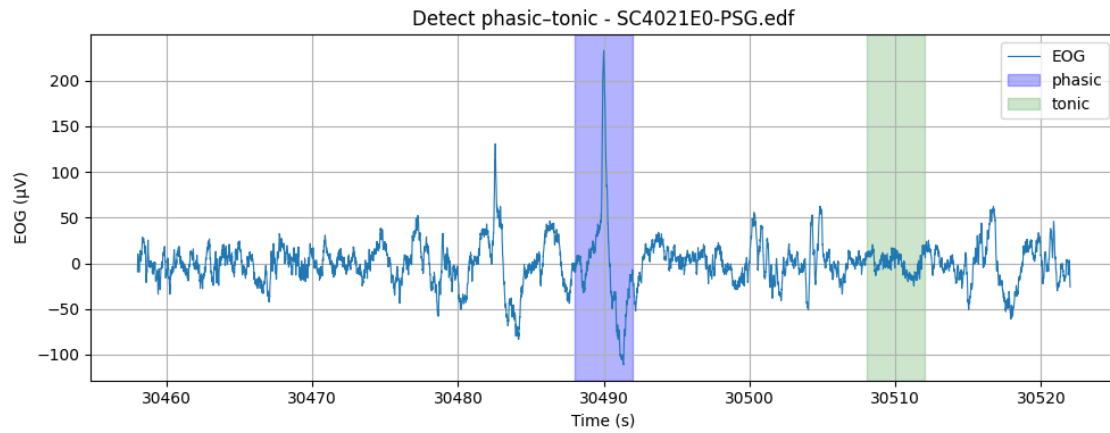
[3] SC4011E0-PSG.edf ...
 phasic windows: 98, tonic windows: 132



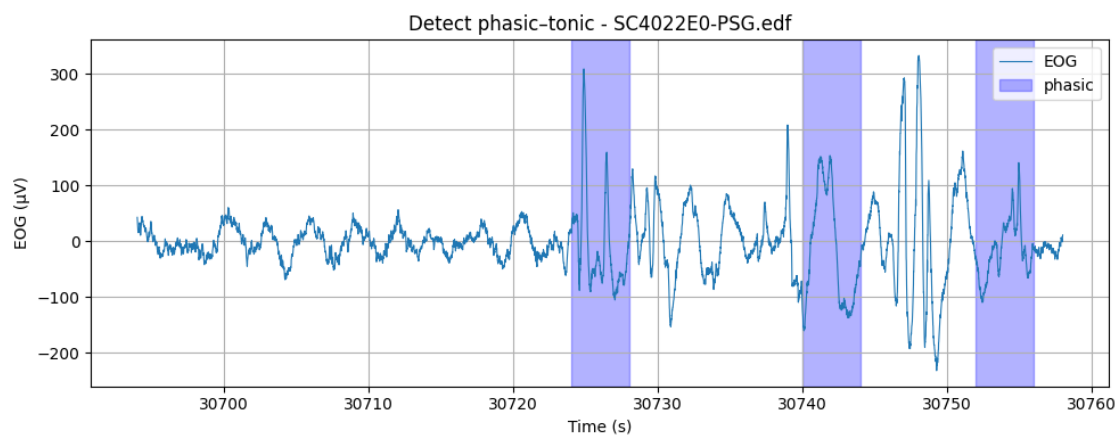
[4] SC4012E0-PSG.edf ...
 phasic windows: 173, tonic windows: 28



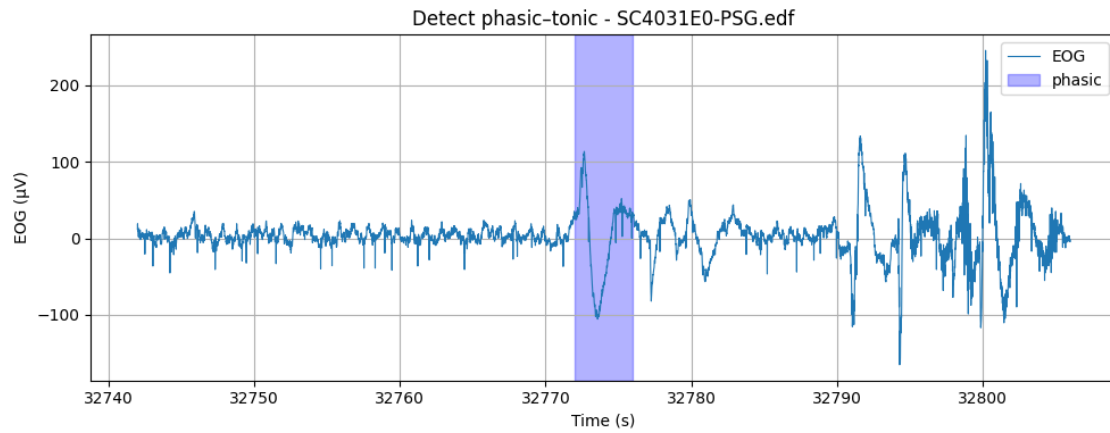
[5] SC4021E0-PSG.edf ...
 phasic windows: 86, tonic windows: 129



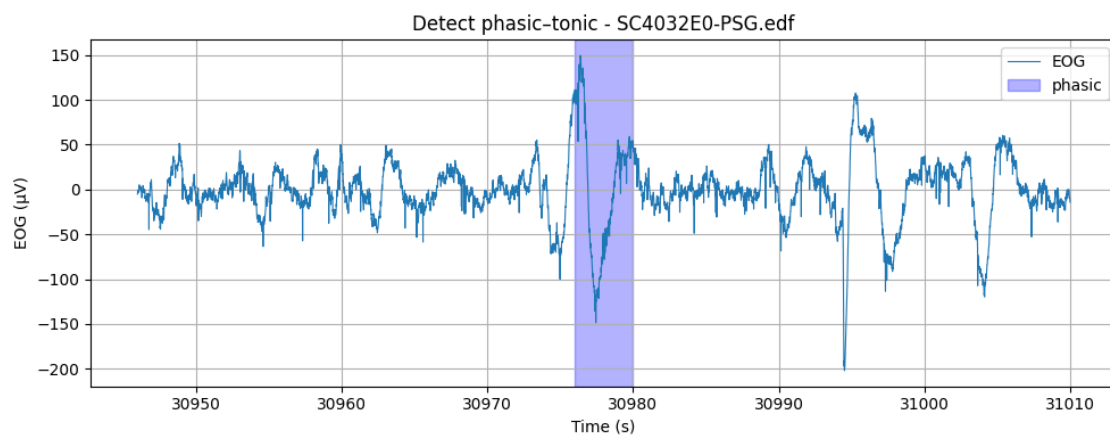
[6] SC4022E0-PSG.edf ...
 phasic windows: 144, tonic windows: 31



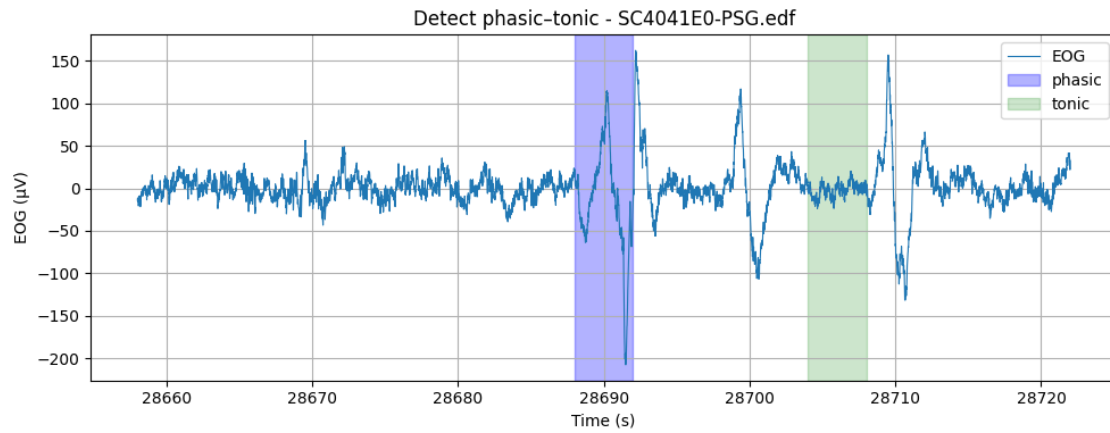
[7] SC4031E0-PSG.edf ...
 phasic windows: 130, tonic windows: 47



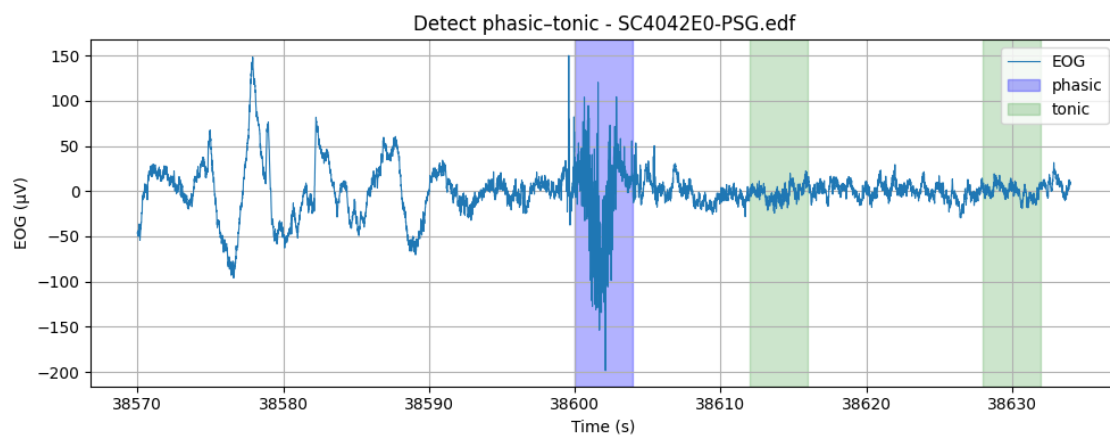
[8] SC4032E0-PSG.edf ...
 phasic windows: 131, tonic windows: 16



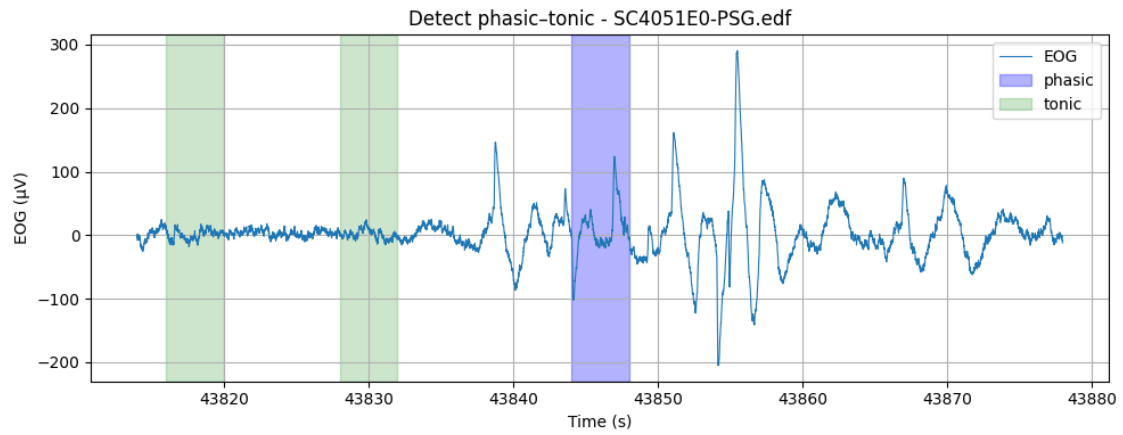
[9] SC4041E0-PSG.edf ...
 phasic windows: 123, tonic windows: 109



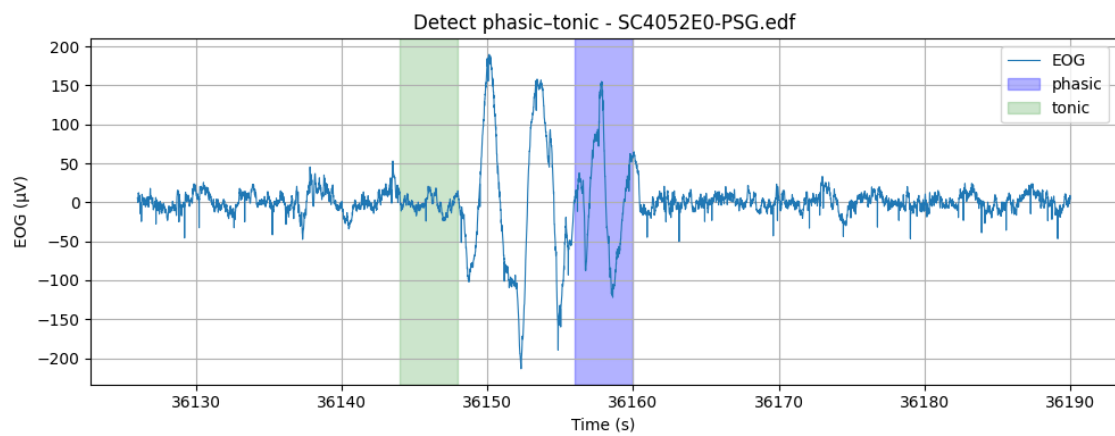
[10] SC4042E0-PSG.edf ...
 phasic windows: 175, tonic windows: 213



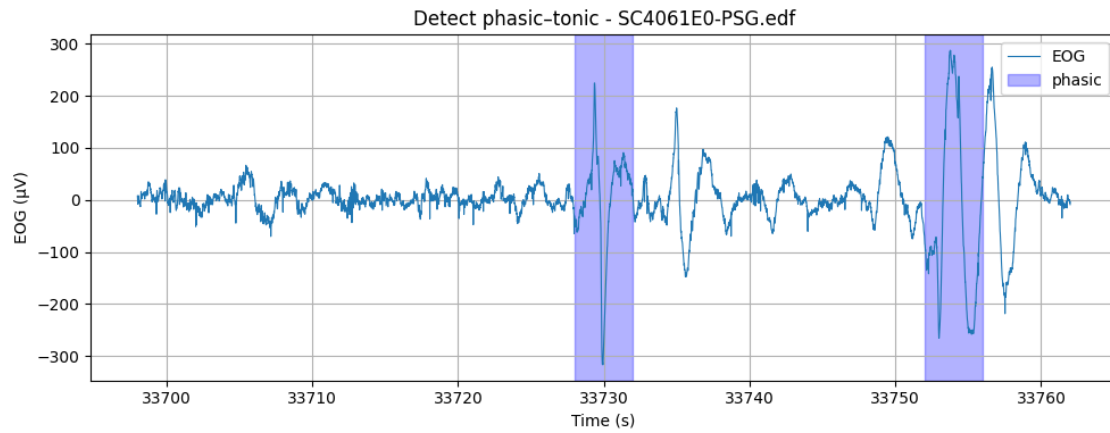
[11] SC4051E0-PSG.edf ...
 phasic windows: 13, tonic windows: 83



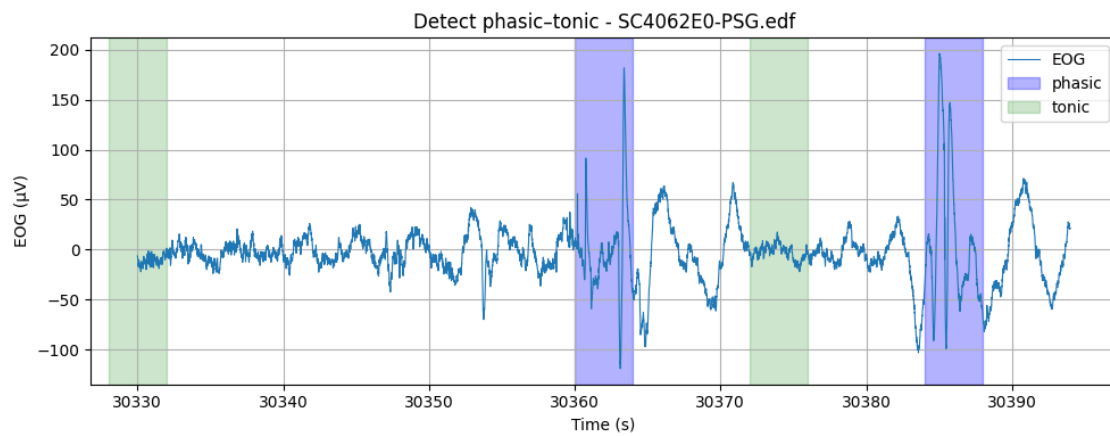
[12] SC4052E0-PSG.edf ...
 phasic windows: 110, tonic windows: 35



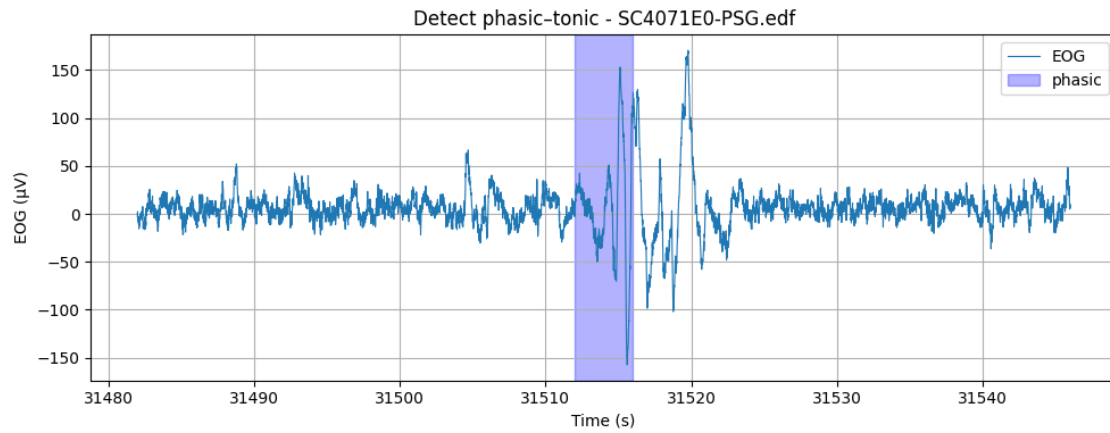
[13] SC4061E0-PSG.edf ...
 phasic windows: 59, tonic windows: 6



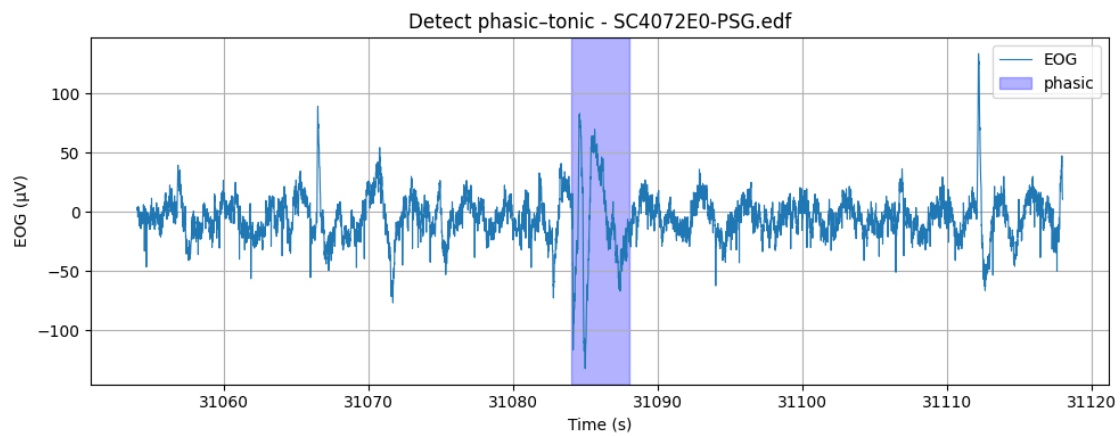
[14] SC4062E0-PSG.edf ...
 phasic windows: 91, tonic windows: 175



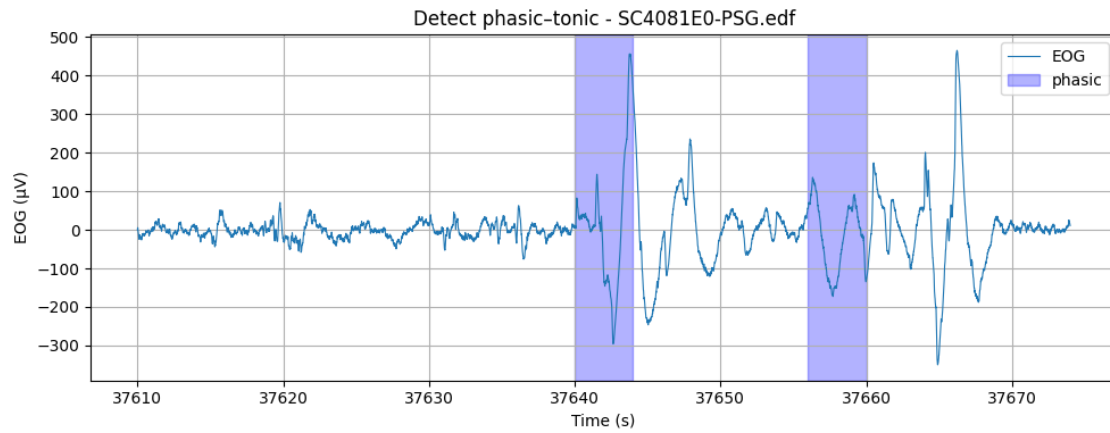
[15] SC4071E0-PSG.edf ...
 phasic windows: 86, tonic windows: 59



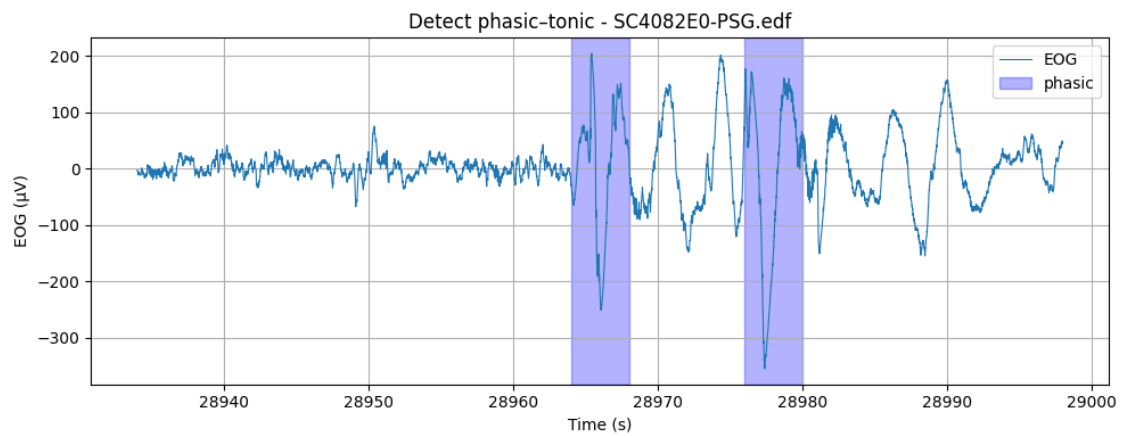
[16] SC4072E0-PSG.edf ...
 phasic windows: 128, tonic windows: 2



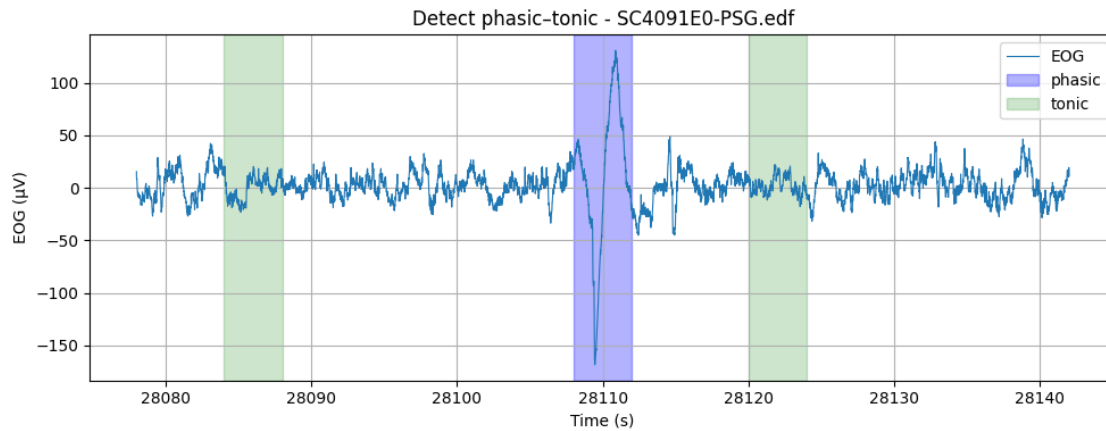
[17] SC4081E0-PSG.edf ...
 phasic windows: 116, tonic windows: 51



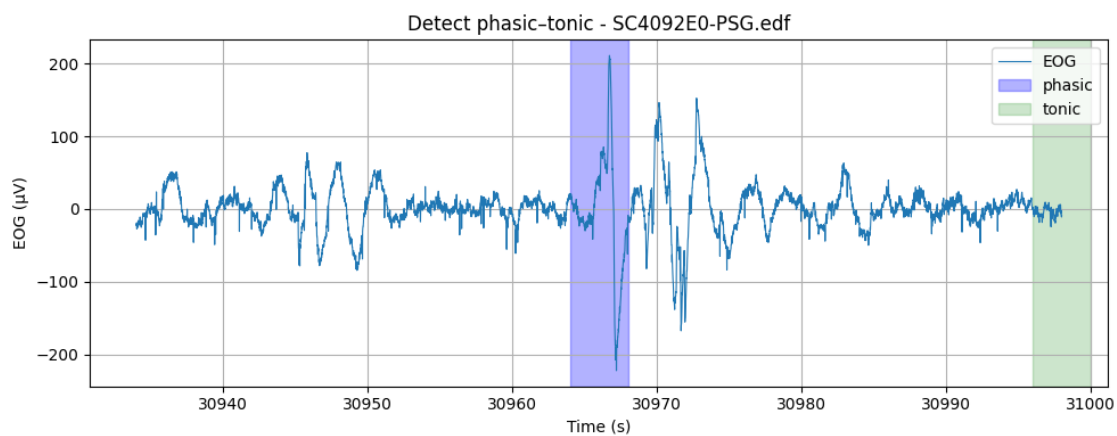
[18] SC4082E0-PSG.edf ...
 phasic windows: 241, tonic windows: 117



[19] SC4091E0-PSG.edf ...
 phasic windows: 78, tonic windows: 207



[20] SC4092E0-PSG.edf ...
 phasic windows: 109, tonic windows: 71



```
[19]: N_subjects=20
for i, (psg_path, hyp_path) in enumerate(zip(psg_file[:N_subjects], hyp_file[:
↪N_subjects]), start=1):

    print(f"[{i}] {os.path.basename(psg_path)} ...")

    raw, eeg, eog, sf = preprocess(psg_path, hyp_path, EEG_CH, EOG_CH)

    rem_mask_bool = rem_mask(raw, rem_label="Sleep stage R")

    phasic, tonic = detect_phasic_tonic(eog, sf, rem_mask_bool)
```

```
# detect phasic and tonic windows
phasic, tonic = detect_phasic_tonic(eog, sf, rem_mask_bool)
print(f"  phasic windows: {len(phasic)}, tonic windows: {len(tonic)}")
```

```
[1] SC4001E0-PSG.edf ...
    phasic windows: 60, tonic windows: 71
[2] SC4002E0-PSG.edf ...
    phasic windows: 89, tonic windows: 12
[3] SC4011E0-PSG.edf ...
    phasic windows: 98, tonic windows: 132
[4] SC4012E0-PSG.edf ...
    phasic windows: 173, tonic windows: 28
[5] SC4021E0-PSG.edf ...
    phasic windows: 86, tonic windows: 129
[6] SC4022E0-PSG.edf ...
    phasic windows: 144, tonic windows: 31
[7] SC4031E0-PSG.edf ...
    phasic windows: 130, tonic windows: 47
[8] SC4032E0-PSG.edf ...
    phasic windows: 131, tonic windows: 16
[9] SC4041E0-PSG.edf ...
    phasic windows: 123, tonic windows: 109
[10] SC4042E0-PSG.edf ...
    phasic windows: 175, tonic windows: 213
[11] SC4051E0-PSG.edf ...
    phasic windows: 13, tonic windows: 83
[12] SC4052E0-PSG.edf ...
    phasic windows: 110, tonic windows: 35
[13] SC4061E0-PSG.edf ...
    phasic windows: 59, tonic windows: 6
[14] SC4062E0-PSG.edf ...
    phasic windows: 91, tonic windows: 175
[15] SC4071E0-PSG.edf ...
    phasic windows: 86, tonic windows: 59
[16] SC4072E0-PSG.edf ...
    phasic windows: 128, tonic windows: 2
[17] SC4081E0-PSG.edf ...
    phasic windows: 116, tonic windows: 51
[18] SC4082E0-PSG.edf ...
    phasic windows: 241, tonic windows: 117
[19] SC4091E0-PSG.edf ...
    phasic windows: 78, tonic windows: 207
[20] SC4092E0-PSG.edf ...
    phasic windows: 109, tonic windows: 71
```

0.0.7 PSD on Phasic and tonic

```
[14]: import os
import random
os.makedirs("PSD_images", exist_ok=True)

N_subjects = 20

# Maximum number of segments used per condition (phasic / tonic)
MAX_SEG = 50

psd_phasic_all = []
psd_tonic_all = []

for i, (psg_path, hyp_path) in enumerate(
    zip(psg_file[:N_subjects], hyp_file[:N_subjects]), start=1):

    print(f"Subject {i:02d}")

    raw, eeg, eog, sf = preprocess(psg_path, hyp_path, EEG_CH, EOG_CH)

    rem_mask_bool = rem_mask(raw)

    phasic, tonic = detect_phasic_tonic(eog, sf, rem_mask_bool)

    n_seg = min(len(phasic), len(tonic), MAX_SEG)

    if n_seg < 20:
        print("  skipped (too few segments)")
        continue

    # Randomly select equal number of phasic and tonic segments
    phasic_sel = random.sample(phasic, n_seg)
    tonic_sel = random.sample(tonic, n_seg)

    # Welch parameters:
    # 2-second windows
    # 50% overlap
    nperseg = int(2 * sf)
    noverlap = int(0.5 * sf)

    psd_phasic = []
```

```

psd_tonic = []

# PSD for phasic
for s, e in phasic_sel:
    seg = eeg[s:e]

    if len(seg) < nperseg:
        continue

    f, p = welch(
        seg,
        fs=sf,
        window='hamming',
        nperseg=nperseg,
        noverlap=nonoverlap
    )
    psd_phasic.append(p)

# PSD for tonic
for s, e in tonic_sel:
    seg = eeg[s:e]

    if len(seg) < nperseg:
        continue

    f, p = welch(
        seg,
        fs=sf,
        window='hamming',
        nperseg=nperseg,
        noverlap=nonoverlap
    )
    psd_tonic.append(p)

if len(psd_phasic) == 0 or len(psd_tonic) == 0:
    continue

psd_phasic = np.mean(psd_phasic, axis=0)
psd_tonic = np.mean(psd_tonic, axis=0)

psd_phasic_all.append(psd_phasic)
psd_tonic_all.append(psd_tonic)

```

```

# ----- Plot & Save -----
plt.figure(figsize=(7, 4))
plt.plot(f, 10 * np.log10(psd_phasic), label='Phasic REM')
plt.plot(f, 10 * np.log10(psd_tonic), label='Tonic REM')

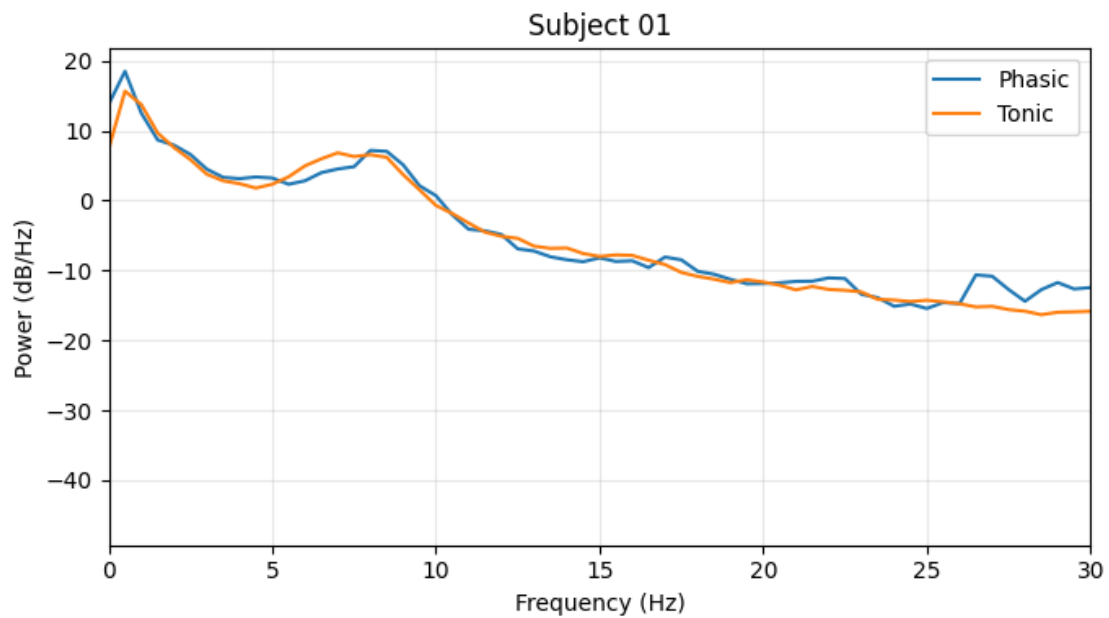
plt.title(f'Subject {i:02d}')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Power Spectral Density (dB/Hz)')
plt.xlim(0, 30)
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()

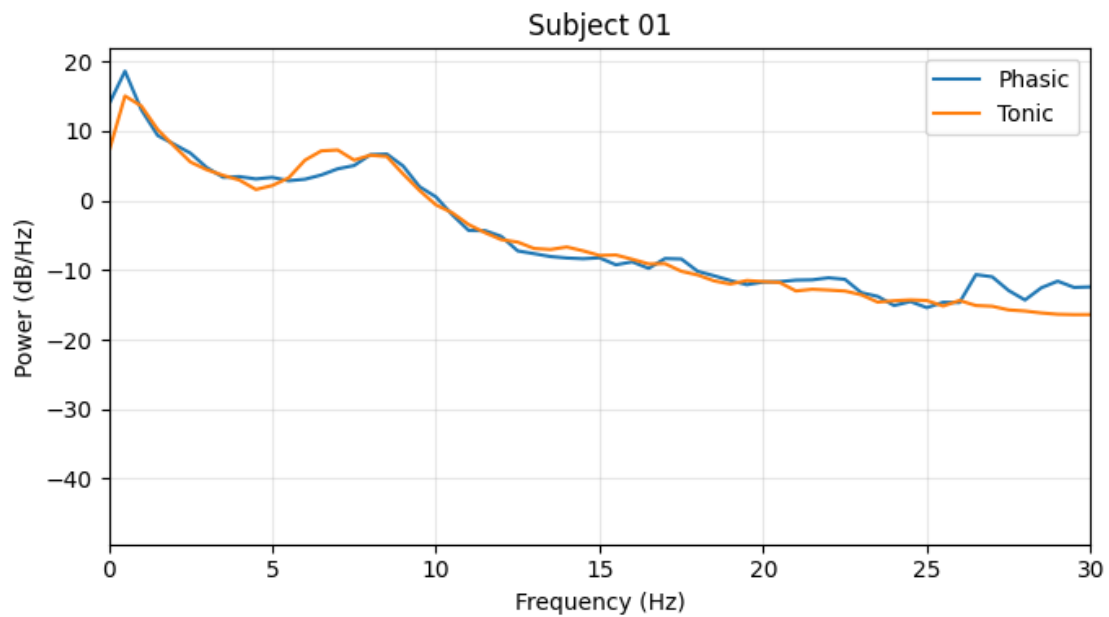
plt.savefig(f"PSD_images/subject_{i}.png", dpi=300, bbox_inches='tight')

plt.show()

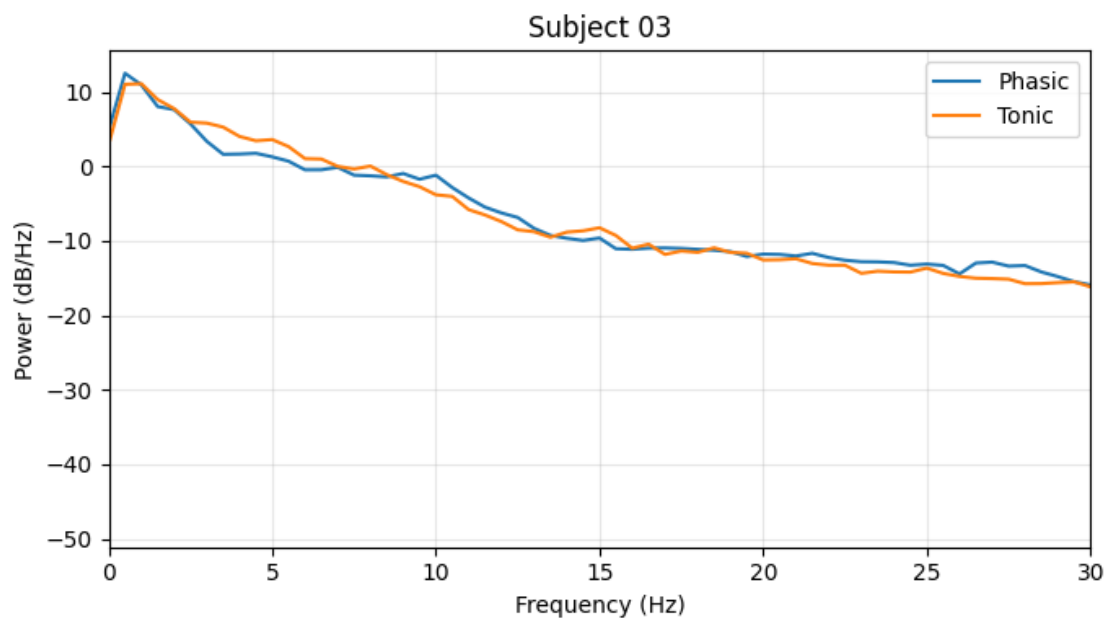
```

Subject 01

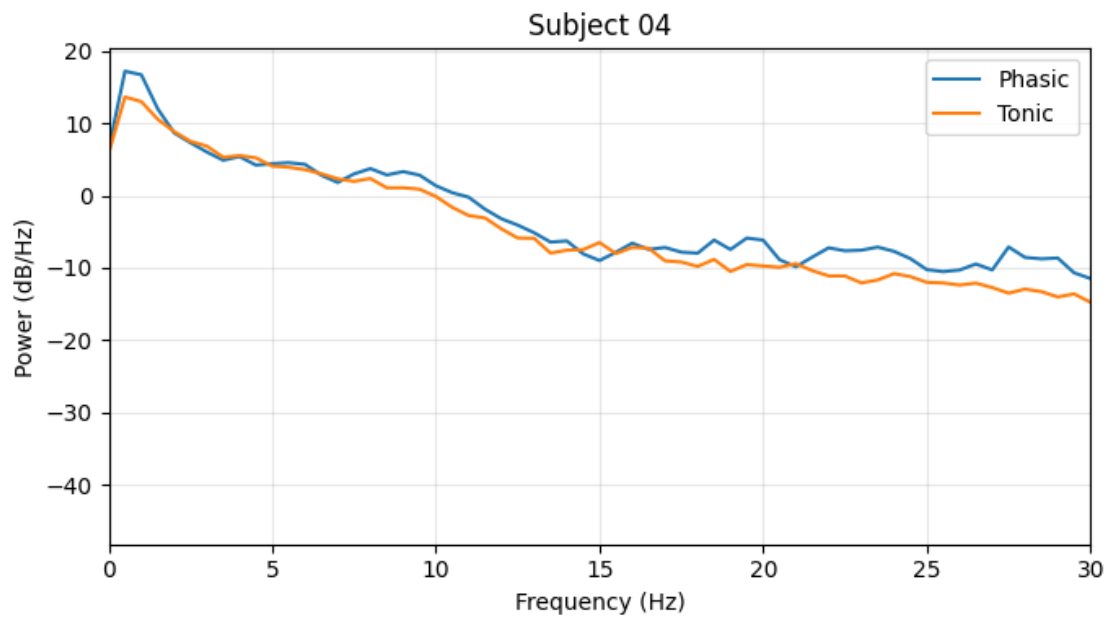




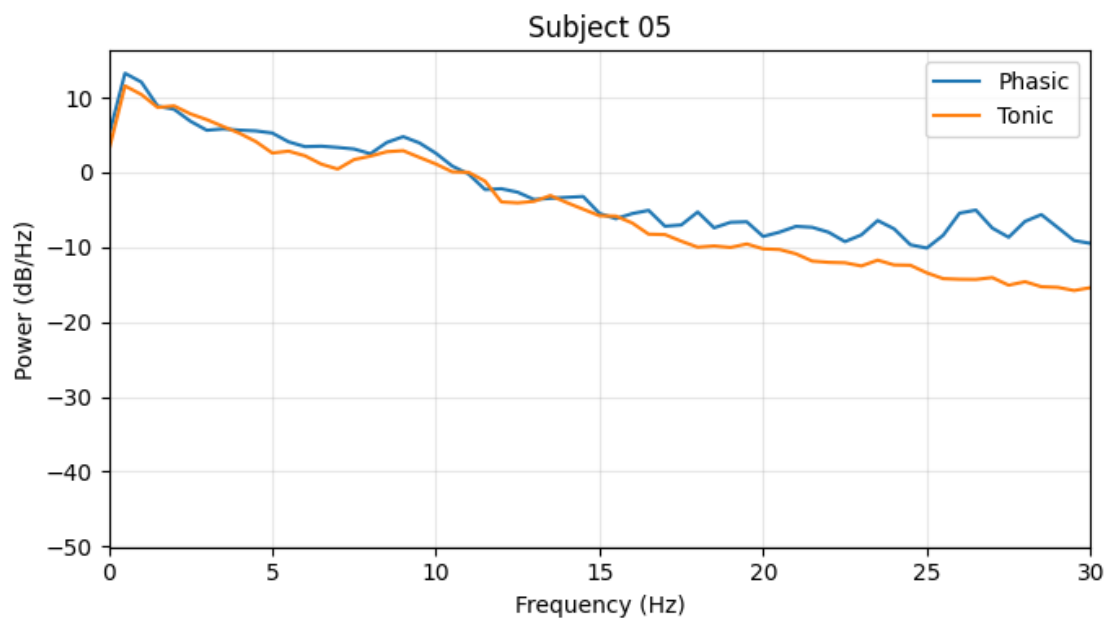
Subject 02
skipped (too few segments)
Subject 03



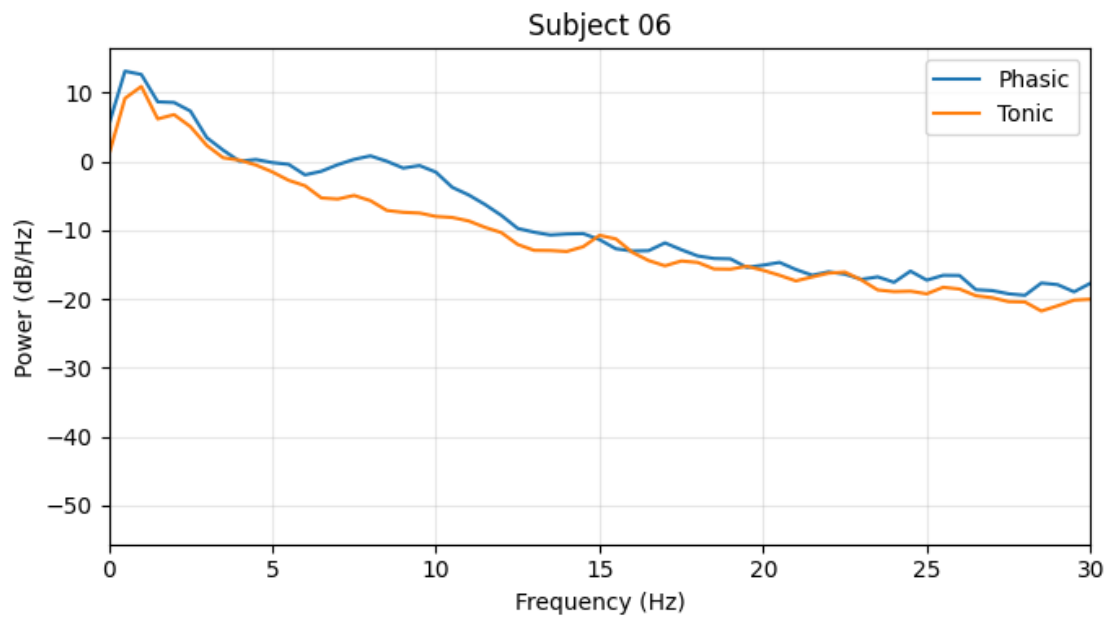
Subject 04



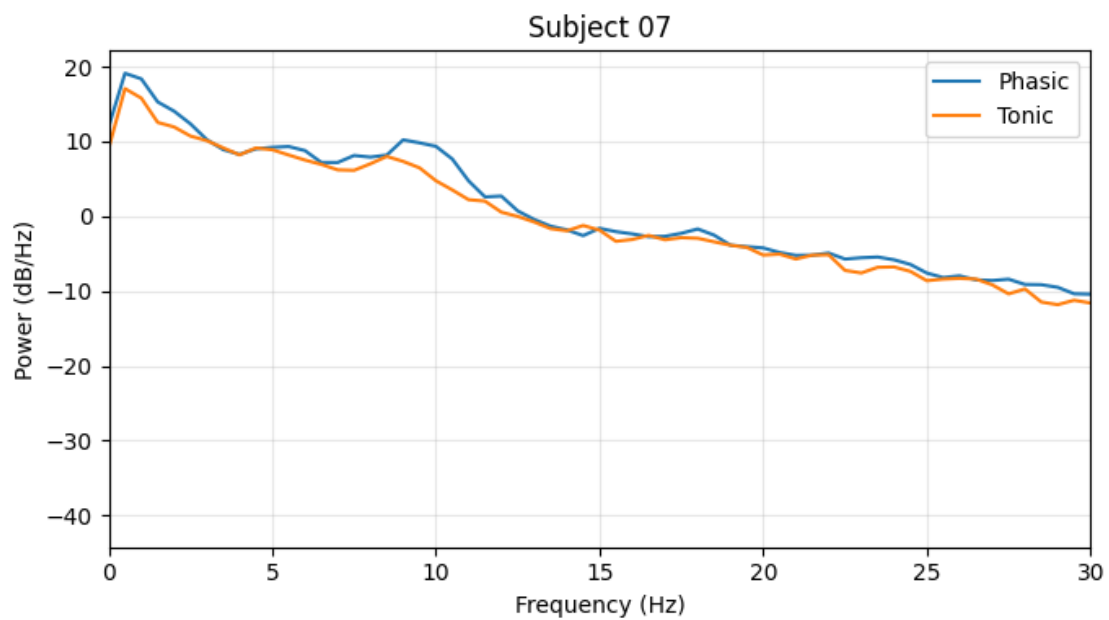
Subject 05



Subject 06



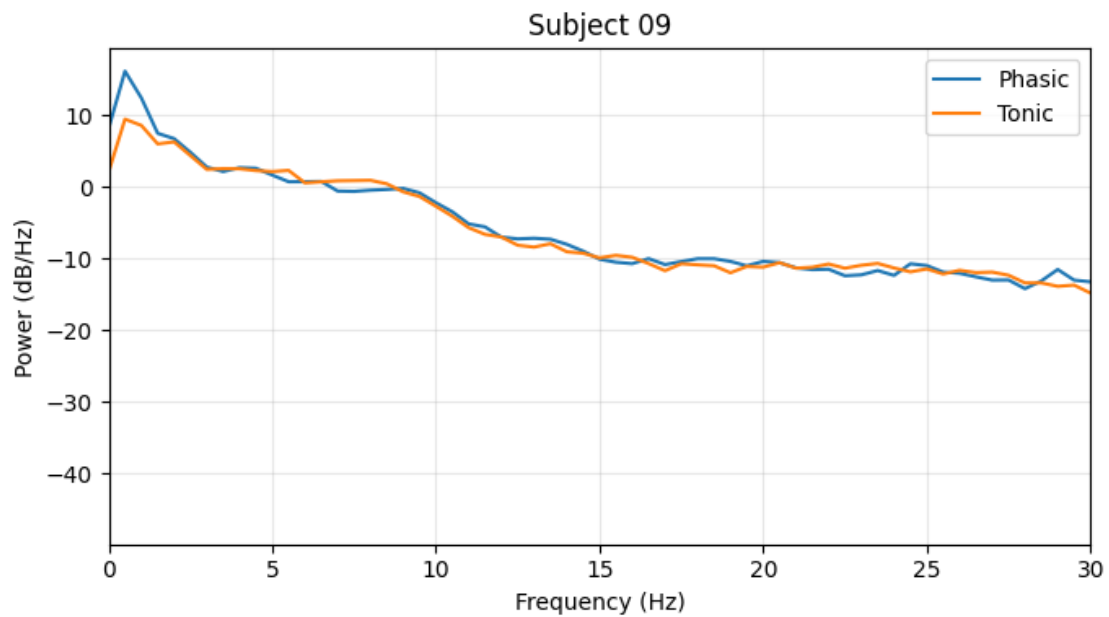
Subject 07



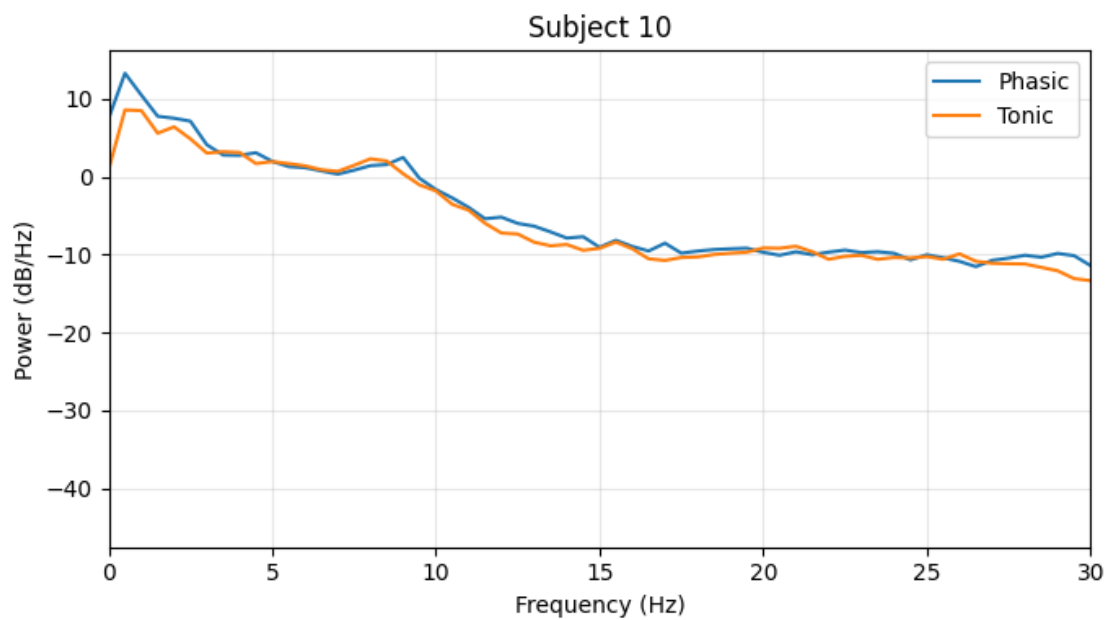
Subject 08

skipped (too few segments)

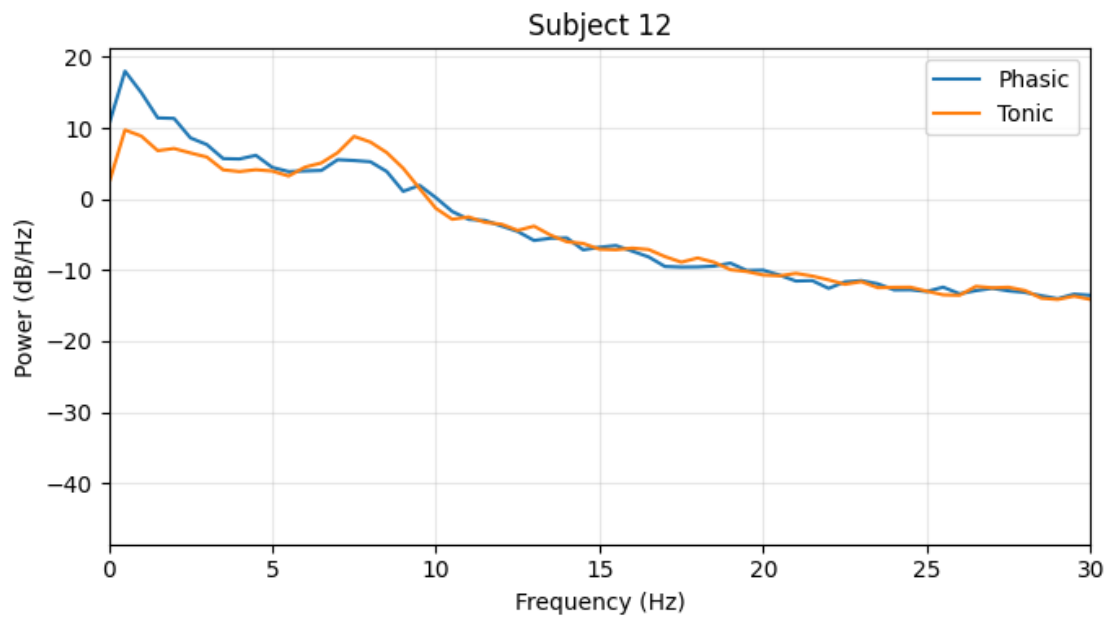
Subject 09



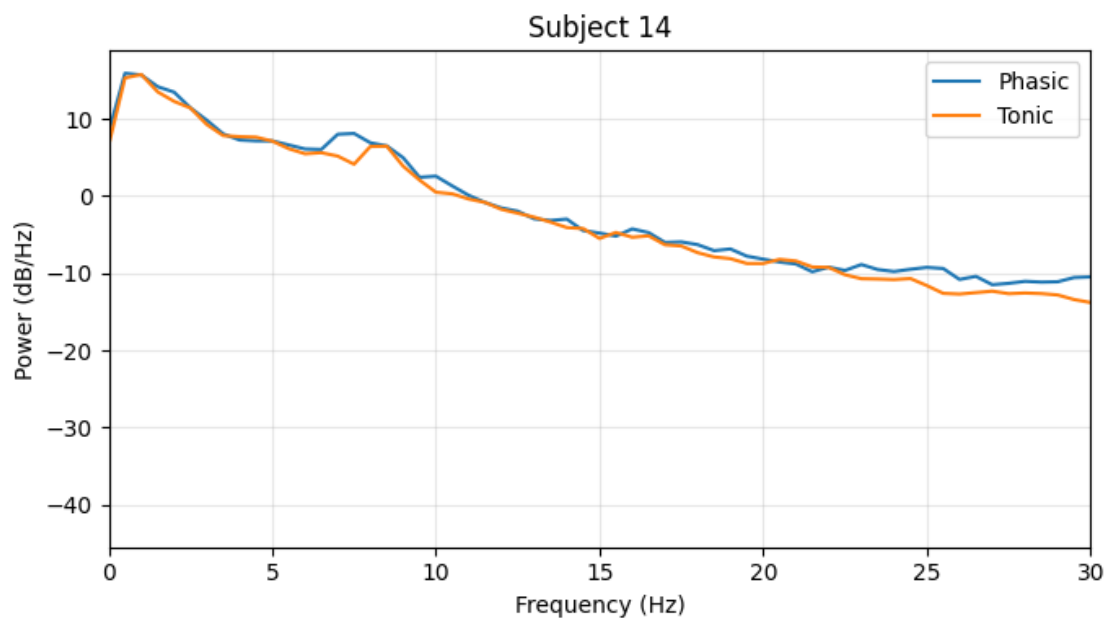
Subject 10



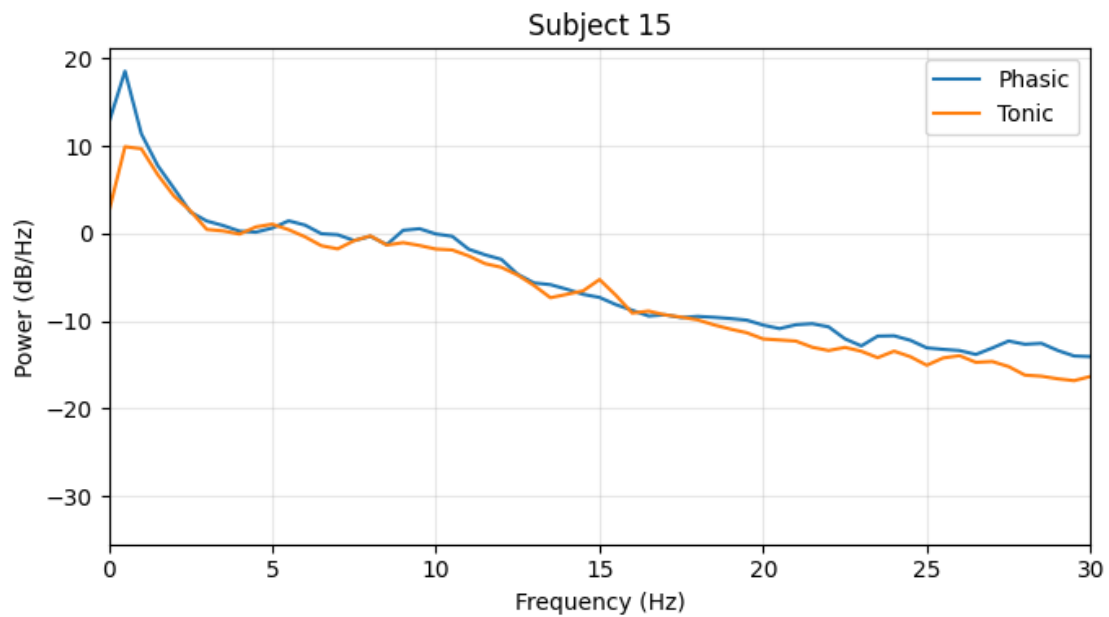
Subject 11
skipped (too few segments)
Subject 12



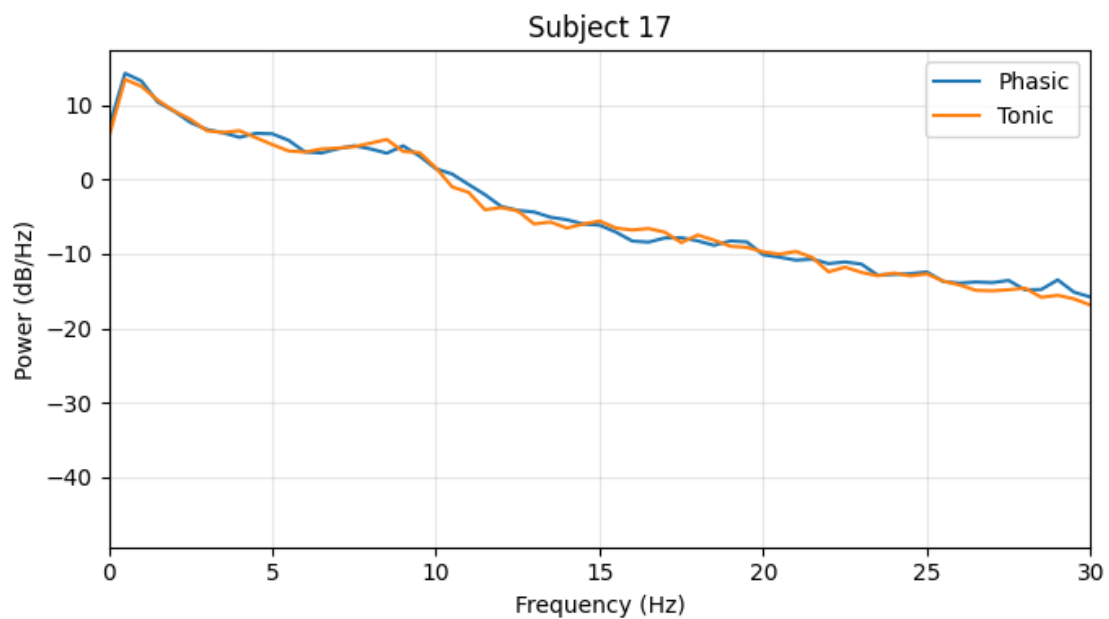
Subject 13
skipped (too few segments)
Subject 14



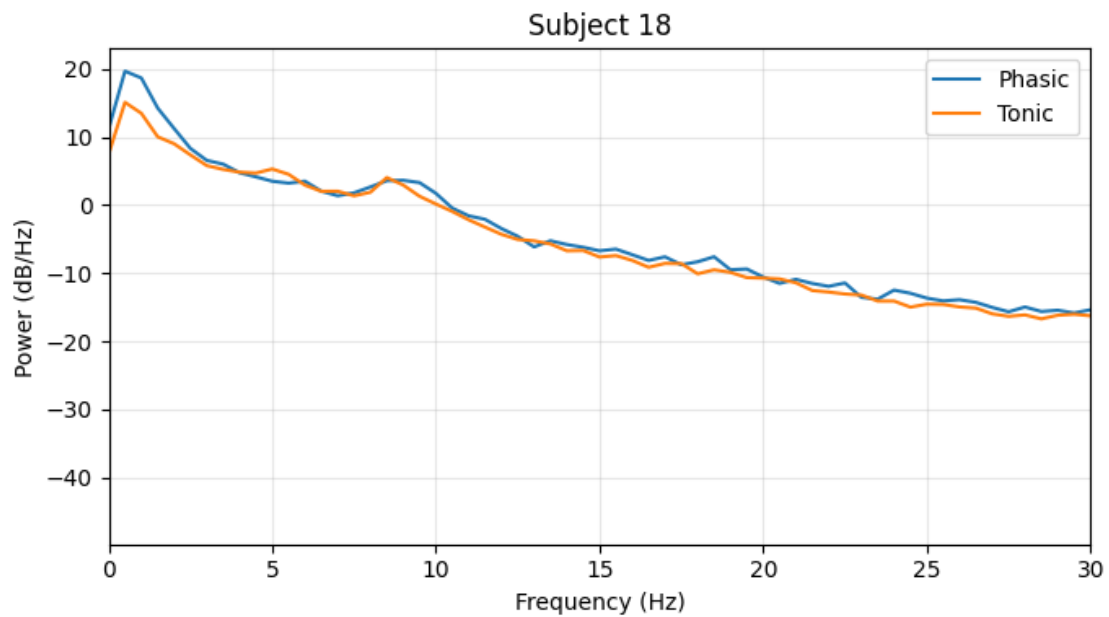
Subject 15



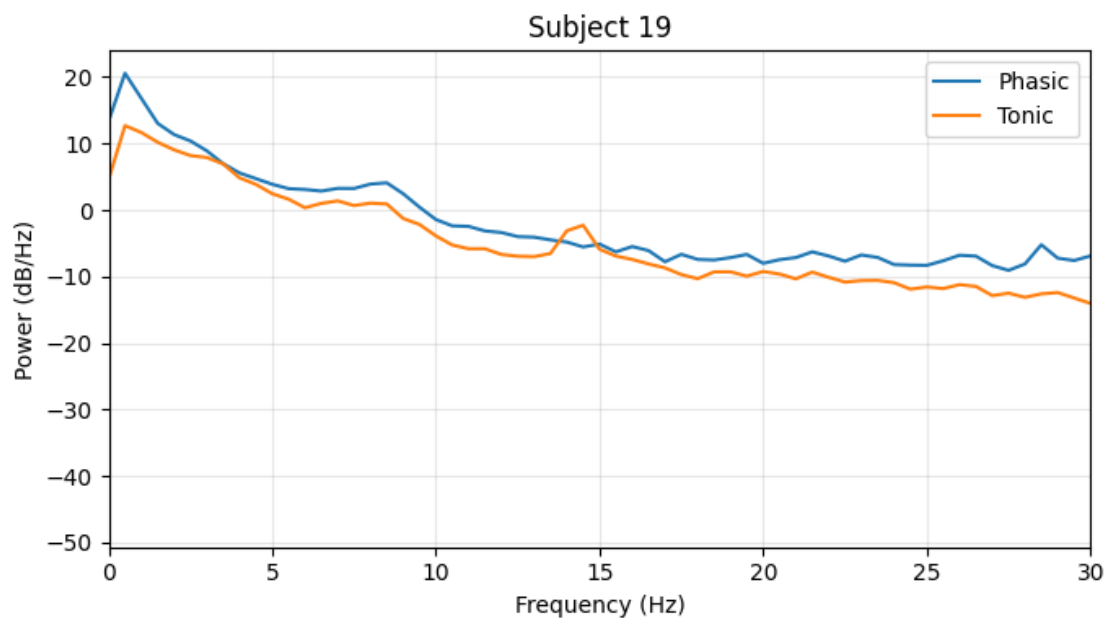
Subject 16
skipped (too few segments)
Subject 17



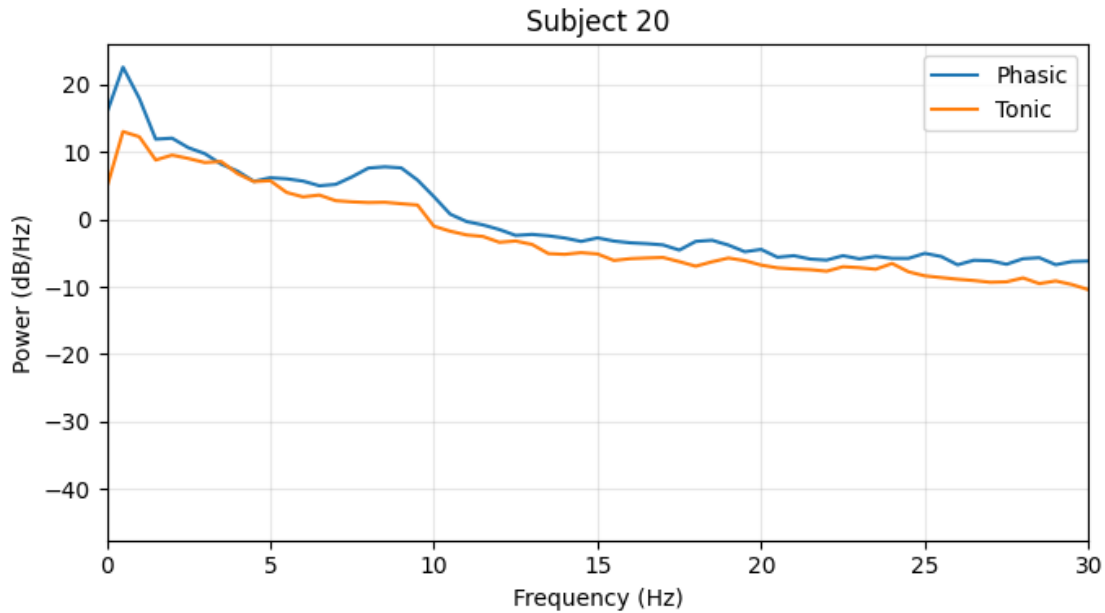
Subject 18



Subject 19



Subject 20



0.0.8 Mean PSD + SEM

```
[15]: import numpy as np

psd_phasic_all = np.array(psd_phasic_all)
psd_tonic_all = np.array(psd_tonic_all)

mean_phasic = np.mean(psd_phasic_all, axis=0)
sem_phasic = np.std(psd_phasic_all, axis=0) / np.sqrt(psd_phasic_all.shape[0])

mean_tonic = np.mean(psd_tonic_all, axis=0)
sem_tonic = np.std(psd_tonic_all, axis=0) / np.sqrt(psd_tonic_all.shape[0])
#Fixed with dB and log scale
mean_phasic_db = 10 * np.log10(mean_phasic)
mean_tonic_db = 10 * np.log10(mean_tonic)

# SEM
sem_phasic_db = 10 * np.log10(mean_phasic + sem_phasic) - mean_phasic_db
sem_tonic_db = 10 * np.log10(mean_tonic + sem_tonic) - mean_tonic_db

plt.figure(figsize=(11,6))
plt.plot(f, mean_phasic_db, lw=2.5, label='Phasic REM')
plt.fill_between(f, mean_phasic_db - sem_phasic_db, mean_phasic_db +
    ↪sem_phasic_db,
                alpha=0.5)
```

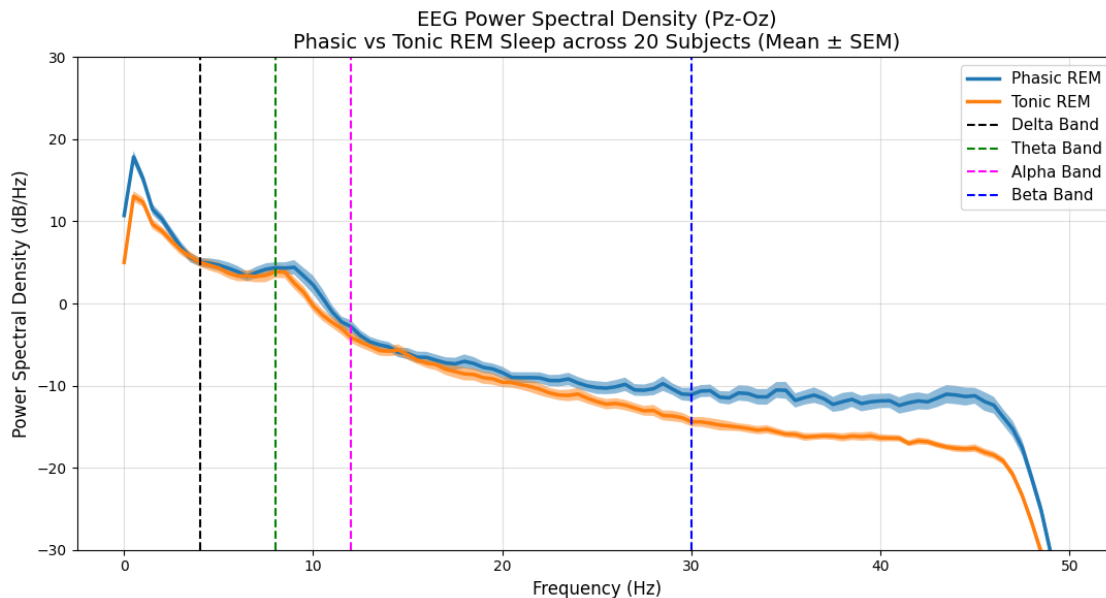
```

plt.plot(f, mean_tonic_db, lw=2.5, label='Tonic REM')
plt.fill_between(f, mean_tonic_db - sem_tonic_db, mean_tonic_db + sem_tonic_db,
    alpha=0.5)

plt.axvline(x=4, color='black', linestyle='--', label='Delta Band')
plt.axvline(x=8, color='green', linestyle='--', label='Theta Band')
plt.axvline(x=12, color='magenta', linestyle='--', label='Alpha Band')
plt.axvline(x=30, color='blue', linestyle='--', label='Beta Band')

plt.xlabel("Frequency (Hz)", fontsize=12)
plt.ylabel("Power Spectral Density (dB/Hz)", fontsize=12)
plt.title("EEG Power Spectral Density (Pz-Oz)\n"
    "Phasic vs Tonic REM Sleep across 20 Subjects (Mean ± SEM)",
    fontsize=14)
plt.legend(fontsize=11)
plt.grid(alpha=0.4)
plt.ylim(-30, 30)
plt.tight_layout()
plt.savefig("FINALAVERAGE_PSD.png", dpi=400, bbox_inches='tight')
plt.show()

```



0.0.9 Bar PLOT

```

[57]: bands = {
    "Delta": (2, 4),
    "Theta": (4, 8),
    "Alpha1": (8, 10),

```



```

    "Alpha2": (10, 12),
    "Beta": (12, 30),
    "Gamma": (30, 48),
}

means_phasic = []
means_tonic = []

for name, (f1, f2) in bands.items():
    idx = np.where((f >= f1) & (f <= f2))
    means_phasic.append(np.mean(psd_phasic[idx]))
    means_tonic.append(np.mean(psd_tonic[idx]))

plt.bar(np.arange(len(bands))-0.15, means_phasic, width=0.3, label="Phasic")
plt.bar(np.arange(len(bands))+0.15, means_tonic, width=0.3, label="Tonic")
plt.xticks(np.arange(len(bands)), list(bands.keys()))
plt.yscale("log")
plt.legend()
plt.show()

```

